

The "inverse" sub-differential

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$

$$f^*(y) \triangleq \sup_x \{y^T x - f(x)\}$$

If f is convex and closed then: $f^{**} = f$

We saw for f a convex quadratic,

∇f and ∇f^* are inverses.

Thm: Let f be closed and convex.

Then ∂f and ∂f^* are inverses (NB: ∂f is in general set-valued)

$$\text{i.e., } y \in \partial f(x) \iff x \in \partial f^*(y) \iff x^T y = f(x) + f^*(y)$$

$$f^*(\hat{y}) \triangleq \sup_x \{ \hat{y}^T x - f(x) \}$$

$$\hat{x}_{\text{opt}} \text{ iff } 0 \in \hat{y} - \partial f(\hat{x})$$

$$s.o.: \quad \hat{y} \in \partial f(\hat{x}) \Rightarrow f^*(\hat{y}) = \hat{y}^T \hat{x} - f(\hat{x})$$

$$\begin{aligned} f^*(y) &= \sup_x \{y^T x - f(x)\} \\ &\geq y^T \hat{x} - f(\hat{x}) \\ &= (y - \hat{y})^T \hat{x} - \underbrace{f(\hat{x})}_{f^*(\hat{y})} + \hat{y}^T \hat{x} \\ &= f^*(\hat{y}) + \underbrace{(y - \hat{y})^T \hat{x}}_{=0} = f^*(\hat{y}) + \hat{x}^T (y - \hat{y}). \end{aligned}$$

$$f^*(y) \stackrel{\downarrow}{\geq} f^*(\hat{y}) + \langle \hat{x}, y - \hat{y} \rangle \iff \hat{x} \in \partial f^*(\hat{y}).$$

So far: $\hat{y} \in \partial f(\hat{z}) \Rightarrow \hat{z} \in \partial f^*(\hat{y})$.

" $\Leftarrow$ " follows by same logic, using $f = f^{**}$.

Dual of a strongly convex function

$$f: \mathbb{R}^n \rightarrow \mathbb{R} \quad \alpha\text{-strongly convex}$$

$$f^*(y) = \sup \{y^T x - f(x)\} \leftarrow \text{unique maximizer } \forall y$$

$$\nabla f^*(y) = \operatorname{argmax} \{y^T x - f(x)\}$$

$$\text{Take } y_1 \in \partial f(x_1), \quad y_2 \in \partial f(x_2)$$

$$\begin{aligned} f \text{ } \alpha\text{-sc: } f(x_1) &\geq f(x_2) + y_2^T (x_1 - x_2) + \frac{\alpha}{2} \|x_1 - x_2\|^2 \\ f(x_2) &\geq f(x_1) + y_1^T (x_2 - x_1) + \frac{\alpha}{2} \|x_1 - x_2\|^2 \end{aligned}$$

$$\alpha \cdot \|x_1 - x_2\|^2 \leq \langle y_1 - y_2, x_1 - x_2 \rangle \leq \|y_1 - y_2\|_* \cdot \|x_1 - x_2\|$$

$$\Rightarrow \|x_1 - x_2\|^2 \leq \frac{1}{\alpha} \|y_1 - y_2\|_*$$

$$\text{i.e., } \|\nabla f^*(y_1) - \nabla f^*(y_2)\| \leq \frac{1}{\alpha} \|y_1 - y_2\|_*$$

$$\Rightarrow f^* \text{ is } \frac{1}{\alpha}\text{-smooth when } f \text{ is } \alpha\text{-sc.}$$

Dual of a smooth convex function

Will show: if f is smooth, then
 f^* is strongly convex

Use: co-coercive:

$$\begin{aligned} \langle \nabla f(x_1) - \nabla f(x_2), x_1 - x_2 \rangle &\geq \frac{1}{\beta} \|\nabla f(x_1) - \nabla f(x_2)\|^2 \\ \left(\begin{array}{l} x_1 \in \partial f^*(y_1), \quad x_2 \in \partial f^*(y_2) \end{array} \right. \\ \left. \right) &\langle y_1 - y_2, x_1 - x_2 \rangle \geq \frac{1}{\beta} \|y_1 - y_2\|^2 \quad (\star) \end{aligned}$$

$$\begin{aligned} \text{Let } h(y) &= f^*(y) - \frac{1}{2\beta} \|y\|^2 \\ \partial h(y) &= \partial f^*(y) - \frac{1}{\beta} y \end{aligned}$$

Check: $(\star) \Rightarrow h$ is monotone $\Rightarrow h$ is convex $\Leftrightarrow f^*$ is strongly convex

Moreau Decomposition

$$x = \text{prox}_f(x) + \text{prox}_{f^*}(x), \quad \forall x$$

Recall: $\text{prox}_f(x) \triangleq \arg \min_u: f(u) + \frac{1}{2} \|u - x\|_2^2$

$$u = \text{prox}_f(x) \Leftrightarrow x - u \in \partial f(u)$$

$$\Leftrightarrow u \in \partial f^*(x - u)$$

$$\Leftrightarrow x - u = \text{prox}_{f^*}(x) \quad (\text{check})$$

Special case: orthogonal decomposition

Check: V be a subspace

$$f = I_V$$

$$\text{M.O.} : x = \text{Proj}_V(x) + \text{Proj}_{V^\perp}(x)$$

Extended Moreau

$$x = \text{prox}_{\lambda f}(x) + \text{prox}_{(\lambda f)^*}(x)$$

$$= \text{prox}_{\lambda f}(x) + \lambda \text{prox}_{f^*/\lambda}(x/\lambda)$$